

MATH2301 – Mathematical Computing

Topic 3: Linear Systems

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Outline

1 Sensitivity Analysis

- Vector and Matrix Norms
- Condition Number
- Stability

2 LU Factorization

- Solving a system by LU factorization
- Existence and uniqueness
- Pivoting
- Instability and partial pivoting
- Computational cost

3 Cholesky Factorization

- Symmetric and positive definite matrices
- Existence and uniqueness
- Computational cost
- Stability

Linear systems arise in

- statistics (linear regression, least squares approximation)

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- solving partial differential equations numerically

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- statistics (linear regression, least squares approximation)
- solving partial differential equations numerically
- signal processing, electrical engineering (electrical networks), chemical engineering (balancing chemical reactions) etc.

A motivational example

Example 1 (Good/bad matrices)

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 + 10^{-10} \end{bmatrix},$$

In solving $A\mathbf{x}_1 = \mathbf{b}$ and $B\mathbf{x}_2 = \mathbf{b}$, which of the two computed solutions $\mathbf{x}_1$ and $\mathbf{x}_2$ is less accurate?

Norms of a vector

To describe the effect of rounding errors on algorithms in linear algebra, we need the following concept.

Definition 2

A **norm** on $\mathbb{R}^n$ is a function that assigns to each $\mathbf{v} \in \mathbb{R}^n$ a real number $\|\mathbf{v}\|$ in such a way that, for all vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and all scalars $\alpha \in \mathbb{R}$,

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- 1 $\|\mathbf{v}\| \geq 0$;
- 2 $\|\mathbf{v}\| = 0$ iff $\mathbf{v} = \mathbf{0}$;
- 3 $\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\|$;
- 4 $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$ (triangle inequality).

The **p -norm** of $\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$ is defined by

$$\|\mathbf{v}\|_p = \left(\sum_{j=1}^n |v_j|^p \right)^{1/p} \quad \text{for } 1 \leq p < \infty,$$

and the **∞ -norm** by

$$\|\mathbf{v}\|_\infty = \max_{1 \leq j \leq n} |v_j| \quad \text{for all } \mathbf{v} \in \mathbb{R}^n.$$

Properties 1–3 in the definition are obvious, but 4 is more difficult.

Example 3

If $\mathbf{v} = (3, -1, 4)$ then what are the values of $\|\mathbf{v}\|_p$ for $p = 1, 2, \infty$?

Notice that the 2-norm is the same as the **Euclidean norm**,

$$\|\mathbf{v}\|_2 = \sqrt{\sum_{j=1}^n (v_j)^2}.$$

The 1-norm is also called the **Manhattan distance**



The Matlab function `norm(x,p)` can be used to compute norms.

Example 4

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 + 10^{-10} \end{bmatrix},$$

Let $\mathbf{b} = (1, 2)^T$ and $\mathbf{b}_1 = \mathbf{b} + (0.01, -0.01)^T$ (so $\mathbf{b}_1$ is a perturbed version of $\mathbf{b}$).

[Go to a Google Form].

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- Compute $\mathbf{x} = A \backslash \mathbf{b}$ and $\mathbf{x}_1 = A \backslash \mathbf{b}_1$.

[Go to a Google Form].

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- Compute $\mathbf{x} = A \backslash \mathbf{b}$ and $\mathbf{x}_1 = A \backslash \mathbf{b}_1$.
- Compute $\mathbf{y} = B \backslash \mathbf{b}$ and $\mathbf{y}_1 = B \backslash \mathbf{b}_1$.

[Go to a Google Form].

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- Compute $\mathbf{x} = A \backslash \mathbf{b}$ and $\mathbf{x}_1 = A \backslash \mathbf{b}_1$.
- Compute $\mathbf{y} = B \backslash \mathbf{b}$ and $\mathbf{y}_1 = B \backslash \mathbf{b}_1$.
- Compare the relative errors

$$\|\mathbf{x}_1 - \mathbf{x}\|_2 / \|\mathbf{x}\|_2 \quad \text{and} \quad \|\mathbf{y}_1 - \mathbf{y}\|_2 / \|\mathbf{y}\|_2.$$

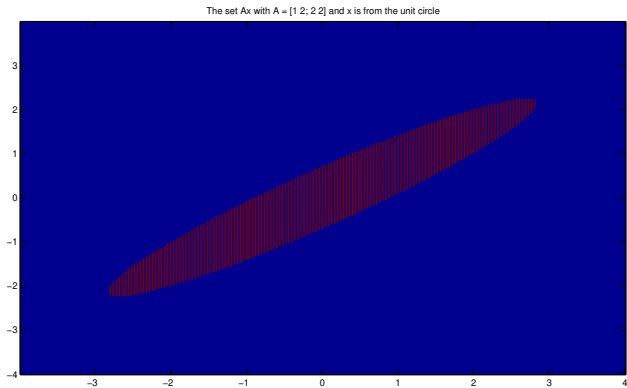
[Go to a Google Form].

Norms of a matrix

Definition 5 (Induced matrix norm)

Given a norm $\|\cdot\|$ on $\mathbb{R}^n$, the **induced matrix norm** is defined by

$$\|A\| = \max_{\|w\|=1} \|Aw\| = \max_{v \neq 0} \frac{\|Av\|}{\|v\|} \quad \text{for } A \in \mathbb{R}^{n \times n}.$$



The set $\{A\mathbf{x} : \|\mathbf{x}\|_2 = 1\}$ with $A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$. The norm $\|A\|_2$ is the radius of the red set.

Theorem 6

The induced matrix norm $\|A\|$ is the smallest constant C such that

$$\|A\mathbf{v}\| \leq C\|\mathbf{v}\| \quad \text{for all } \mathbf{v} \in \mathbb{R}^n.$$

Theorem 7

Any induced norm on $\mathbb{R}^{n \times n}$ has the properties

1. $\|A\| \geq 0$;

2. $\|A\| = 0$ iff $A = 0$;

3. $\|\alpha A\| = |\alpha| \|A\|$;

4. $\|A + B\| \leq \|A\| + \|B\|$.

We denote by $\|A\|_p$ the matrix norm induced by the p -norm on $\mathbb{R}^n$. In practice, $\|A\|_p$ is difficult to compute unless $p = 1$ or ∞ .

Theorem 8

For $A \in \mathbb{R}^{n \times n}$,

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}| = \text{maximum absolute column sum},$$

$$\|A\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}| = \text{maximum absolute row sum}.$$

Example 9

If

$$A = \begin{bmatrix} 1 & 3 & -4 \\ 2 & 0 & -1 \\ 6 & 9 & -1 \end{bmatrix}$$

the what are the values of $\|A\|_1$ and $\|A\|_\infty$?

Condition number

Definition 10

Given a norm on $\mathbb{R}^n$, we define the **condition number** of the matrix A by

$$\kappa(A) = \|A\| \|A^{-1}\|.$$

We define $\kappa(A) = \infty$ whenever A is singular.

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Example 11

Let

$$A = \begin{bmatrix} 1 & -1 \\ k & -1 \end{bmatrix}$$

for some $k > 1$. Compute A^{-1} and then $\kappa_1(A)$?

Definition 12 (Ill-conditioned matrices)

If $\kappa(A)$ is large, then A is said to be an **ill-conditioned** matrix.

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Ill-conditionness of a matrix is a norm-dependent property. However, any two condition numbers are equivalent.

Lemma 13

If $A \in \mathbb{R}^{n \times n}$ then

$$\frac{1}{n} \kappa_2(A) \leq \kappa_1(A) \leq n \kappa_2(A)$$

$$\frac{1}{n} \kappa_\infty(A) \leq \kappa_2(A) \leq n \kappa_\infty(A)$$

$$\frac{1}{n^2} \kappa_1(A) \leq \kappa_\infty(A) \leq n^2 \kappa_1(A).$$

Proof: Exercise!

Lemma 14

Let $A, B \in \mathbb{R}^{n \times n}$ and let $I = [\delta_{ij}]$ denote the identity matrix in $\mathbb{R}^{n \times n}$. Show that for any induced norm $\|\cdot\|$ on $\mathbb{R}^{n \times n}$,

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- 5 $\kappa(A) \geq 1$.

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Proof: Exercise!

Condition number κ_2 of a real symmetric matrix

Definition 15 (Eigenvalues)

Let $A \in \mathbb{R}^{n \times n}$ be an $n \times n$ matrix. A number $\lambda \in \mathbb{C}$ is called an **eigenvalue** of A if there exists a nonzero vector $\mathbf{v} \in \mathbb{C}^n$ such that

$$A\mathbf{v} = \lambda\mathbf{v}$$

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Theorem 16

If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of an $n \times n$ **real** and **symmetric** matrix A such that

$$|\lambda_1| \geq \dots \geq |\lambda_n| > 0$$

then

$$\kappa_2(A) = \frac{|\lambda_1|}{|\lambda_n|}.$$

Example 17

A **symmetric** matrix $A \in \mathbb{R}^{n \times n}$ is known **exactly** and has ordered eigenvalues

$$1004.2, \quad 999.8, \quad \dots, \quad 0.0034, \quad 0.0005.$$

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- 1 Explain why A is nonsingular
- 2 Estimate the condition number of A

Stability

Let $A \in \mathbb{R}^{n \times n}$ be a non-singular matrix and consider the linear system

$$A\mathbf{x} = \mathbf{b}.$$

We now discuss the sensitivity of the solution $\mathbf{x}$ to small perturbations in A and $\mathbf{b}$. Let $\delta\mathbf{x}$ denote the change in $\mathbf{x}$ when A and $\mathbf{b}$ are changed by δA and $\delta\mathbf{b}$, respectively, so that

$$(A + \delta A)(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b}.$$

The following theorem shows that if A is ill-conditioned then the change in $\mathbf{x}$, namely $\delta\mathbf{x}$, may be large even though the perturbations δA and $\delta\mathbf{b}$ are small.

Error analysis

Theorem 18

Suppose that

$$\frac{\|\delta A\|_p}{\|A\|_p} \leq \gamma < \frac{1}{\kappa_p(A)}.$$

Then $A + \delta A$ is invertible. Moreover, if $A\mathbf{x} = \mathbf{b}$ and $(A + \delta A)(\mathbf{x} + \delta \mathbf{x}) = \mathbf{b} + \delta \mathbf{b}$ then

$$\frac{\|\delta \mathbf{x}\|_p}{\|\mathbf{x}\|_p} \leq \frac{\kappa_p(A)}{1 - \gamma \kappa_p(A)} \left(\frac{\|\delta A\|_p}{\|A\|_p} + \frac{\|\delta \mathbf{b}\|_p}{\|\mathbf{b}\|_p} \right).$$

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Conclusions

- Relative error in solution $\preceq \kappa_p(A) \times$ relative error in data
- In IEEE double precision (ϵ of order 10^{-15}), if $\kappa(A)$ is of order 10^k then expect about $15 - k$ correct digits in $\mathbf{x}_*$.

Example 19

Let

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- What is the limiting behaviour of $\kappa_1(A)$ when $\mu \rightarrow 0$?

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Proof

Example 20

Consider the matrix in Example 19 with $\mu = 0.1$. Solve $A\mathbf{x}_1 = \mathbf{b}$, $(A + \delta A)\mathbf{x}_2 = \mathbf{b}$, and $(A + \delta A)\mathbf{x}_3 = \mathbf{b} + \delta\mathbf{b}$ with

$$\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \delta A = \begin{bmatrix} 0 & 0 \\ 0 & 0.1 \end{bmatrix} \quad \delta\mathbf{b} = \begin{bmatrix} 0 \\ 0.1 \end{bmatrix}.$$

The equations to be solved are

$$\begin{bmatrix} 2 & 1 \\ 4 & 2.1 \end{bmatrix} \mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 2 & 1 \\ 4 & 2.2 \end{bmatrix} \mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 2 & 1 \\ 4 & 2.2 \end{bmatrix} \mathbf{x}_3 = \begin{bmatrix} 1 \\ 1.1 \end{bmatrix}.$$

We have (check it!)

$$\mathbf{x}_1 = \begin{bmatrix} 5.5 \\ -10 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 3 \\ -5 \end{bmatrix} \quad \text{and} \quad \mathbf{x}_3 = \begin{bmatrix} 2.7500 \\ -4.5000 \end{bmatrix},$$

Example 20 (cont.)

so that

$$\frac{\|\mathbf{x}_1 - \mathbf{x}_2\|_1}{\|\mathbf{x}_1\|_1} = \frac{7.5}{15.5} = 0.4839 \quad \text{and} \quad \frac{\|\mathbf{x}_1 - \mathbf{x}_3\|_1}{\|\mathbf{x}_1\|_1} = \frac{8.2500}{15.5} = 0.5323.$$

On the other hand,

$$\frac{\|\delta \mathbf{A}\|_1}{\|\mathbf{A}\|_1} = \frac{0.1}{6} = 0.0167 \quad \text{and} \quad \frac{\|\delta \mathbf{b}\|_1}{\|\mathbf{b}\|_1} = \frac{0.1}{2} = 0.05.$$

Hence

$$\frac{\|\mathbf{x}_1 - \mathbf{x}_2\|_1}{\|\mathbf{x}_1\|_1} \approx 29 \frac{\|\delta \mathbf{A}\|_1}{\|\mathbf{A}\|_1} \quad \text{and} \quad \frac{\|\mathbf{x}_1 - \mathbf{x}_3\|_1}{\|\mathbf{x}_1\|_1} \approx 8 \left(\frac{\|\delta \mathbf{A}\|_1}{\|\mathbf{A}\|_1} + \frac{\|\delta \mathbf{b}\|_1}{\|\mathbf{b}\|_1} \right).$$

This example supports Theorem 18:

Relative error in solution $\preceq \kappa_1(\mathbf{A}) \times$ relative error in data

because $\kappa_1(\mathbf{A}) = 183$ according to Example 19.

Preconditioning ill-posed systems

- Theorem 18 shows that when the matrix A is ill-conditioned, solving $A\mathbf{x} = \mathbf{b}$ doesn't give accurate solutions.

Preconditioning ill-posed systems

- Theorem 18 shows that when the matrix A is ill-conditioned, solving $A\mathbf{x} = \mathbf{b}$ doesn't give accurate solutions.
- A remedy is to **precondition** the matrix A . Instead of solving $A\mathbf{x} = \mathbf{b}$ we solve $MA\mathbf{x} = M\mathbf{b}$ where M is an invertible matrix such that $\kappa(MA)$ is much smaller than $\kappa(A)$. We will not discuss this topic further.

Do a small quiz in a Google Form.

A motivational example

Example 21

Suppose the matrix A is given. There are 10^5 input vectors $\mathbf{b}_k$. What is the most effective way to solve 10^5 linear systems $A\mathbf{x}_k = \mathbf{b}_k$?

Definition 22 (Lower triangular matrices)

A matrix $L = [\ell_{ij}] \in \mathbb{R}^{n \times n}$ is said to be **lower triangular** if

$$\ell_{ij} = 0 \quad \text{for } 1 \leq i < j \leq n.$$

We say that L is **unit lower triangular** if, in addition,

$$\ell_{ii} = 1 \quad \text{for } 1 \leq i \leq n.$$

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Example 23

A 4×4 unit lower triangular matrix has the form

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \ell_{21} & 1 & 0 & 0 \\ \ell_{31} & \ell_{32} & 1 & 0 \\ \ell_{41} & \ell_{42} & \ell_{43} & 1 \end{bmatrix}.$$

A unit lower triangular system of equations is easily solved using **forward elimination**.

A unit lower triangular system of equations is easily solved using **forward elimination**.

Example 24

To solve

$$\begin{aligned}x_1 &= 2, \\-2x_1 + x_2 &= -5, \\x_1 + 4x_2 + x_3 &= -2, \\3x_1 + 3x_3 + x_4 &= 10,\end{aligned}$$

we do

$$\begin{aligned}x_1 &= 2, \\x_2 &= -5 + 2x_1 = -5 + 2 \times 2 = -1, \\x_3 &= -2 - x_1 - 4x_2 = -2 - 2 - 4 \times (-1) = 0, \\x_4 &= 10 - 3x_1 - 3x_3 = 10 - 3 \times 2 - 3 \times 0 = 4.\end{aligned}$$

Definition 25 (Upper triangular matrices)

A matrix $U = [u_{ij}] \in \mathbb{R}^{n \times n}$ is said to be **upper triangular** if

$$u_{ij} = 0 \quad \text{for } 1 \leq j < i \leq n,$$

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Example 26

A 4×4 upper triangular matrix has the form

$$U = \begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & u_{22} & u_{23} & u_{24} \\ 0 & 0 & u_{33} & u_{34} \\ 0 & 0 & 0 & u_{44} \end{bmatrix}.$$

Definition 25 (Upper triangular matrices)

A matrix $U = [u_{ij}] \in \mathbb{R}^{n \times n}$ is said to be **upper triangular** if

$$u_{ij} = 0 \quad \text{for } 1 \leq j < i \leq n,$$

and **unit upper triangular** if, in addition, $u_{ii} = 1$ for all i .

Example 26

A 4×4 upper triangular matrix has the form

$$U = \begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & u_{22} & u_{23} & u_{24} \\ 0 & 0 & u_{33} & u_{32} \\ 0 & 0 & 0 & u_{44} \end{bmatrix}.$$

An upper triangular linear system is easily solved using **back substitution**, provided every diagonal entry is non-zero, i.e., $u_{ii} \neq 0$ for $1 \leq i \leq n$.

Definition 27 (LU-factorization)

An **LU-factorization** of a matrix $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ is a representation of the form

$$A = LU,$$

for some **unit** lower triangular matrix L and some upper triangular matrix U .

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is easy:

- First solve $L\mathbf{y} = \mathbf{b}$ using forward elimination,
- Then solve $U\mathbf{x} = \mathbf{y}$ using back substitution.
- In this way,

$$A\mathbf{x} = (LU)\mathbf{x} = L(U\mathbf{x}) = L\mathbf{y} = \mathbf{b},$$

as required.

Example 28

Check that

$$\begin{bmatrix} 1 & 0 & 3 \\ 2 & 2 & 2 \\ 3 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 2 & -4 \\ 0 & 0 & 7 \end{bmatrix}$$

and then solve

$$\begin{bmatrix} 1 & 0 & 3 \\ 2 & 2 & 2 \\ 3 & 6 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 7 \end{bmatrix}.$$

To find the LU factorization: Gaussian elimination

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- 1 Set $a_{ij}^{(1)} = a_{ij}$ for $i, j = 1, \dots, n$;

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2 For $k = 1, \dots, n - 1$ do

for $i = k + 1, \dots, n$

$$\ell_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$$

for $j = k + 1, \dots, n$

$$a_{ij}^{(k+1)} = a_{ij}^{(k)} - \ell_{ik} a_{kj}^{(k)}$$

end do

end do

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for $j = k + 1, \dots, n$

$$a_{ij}^{(k+1)} = a_{ij}^{(k)} - \ell_{ik} a_{kj}^{(k)}$$

end do

end do

3 At the end of the procedure, the entries of L are known, whereas the entries of U are

$$u_{ij} = a_{ij}^{(i)} \quad \text{for } i = 1, \dots, n \text{ and } j = i, \dots, n.$$

The elements $a_{kk}^{(k)}$ must all be different from zero and are called **pivot elements**.

Example 29

Find the LU-factorization of

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 2 & 2 \\ 3 & 6 & 4 \end{bmatrix}.$$

Example 30

Show that the following matrix A is non-singular but does not have an LU factorization

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 2 & 2 \\ 3 & 6 & 4 \end{bmatrix}.$$

Principal submatrices

Definition 31 (Principal submatrix)

Let A be an $n \times n$ matrix. For $i = 1, 2, \dots, n$, the **principal submatrix** of order i of A is an $i \times i$ matrix whose rows and columns are the first i rows and columns of A .

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Example 32

Let

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 2 & 2 \\ 3 & 6 & 4 \end{bmatrix}.$$

Then

$$A_1 = [1], \quad A_2 = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, \quad A_3 = A.$$

Existence and uniqueness of LU factorization

Theorem 33

*For a given matrix A of size $n \times n$, its LU factorization exists and is unique if and only if all **principal submatrices** A_i of A , for $i = 1, 2, \dots, n - 1$, are non-singular.*

Existence and uniqueness of LU factorization

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Example 34

Use Theorem 33 to show that the matrix

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 2 & 2 \\ 3 & 6 & 4 \end{bmatrix}$$

does not have an LU factorization.

Example 35

Show that the following matrix has an LU factorization and find this factorization.

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 2 & 2 \\ 3 & 6 & -3 \end{bmatrix}.$$

Note that A is singular.

Pivoting

- Gaussian elimination algorithm requires that all **pivot elements** are nonzero.

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- Therefore, instead of solving $A\mathbf{x} = \mathbf{b}$ we solve $PA\mathbf{x} = P\mathbf{b}$, or $LU\mathbf{x} = P\mathbf{b}$.

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- Therefore, instead of solving $A\mathbf{x} = \mathbf{b}$ we solve $PA\mathbf{x} = P\mathbf{b}$, or $LU\mathbf{x} = P\mathbf{b}$.
- This means we solve the triangular systems $L\mathbf{y} = P\mathbf{b}$ and $U\mathbf{x} = \mathbf{y}$.

Example 36

Use pivoting to find the LU factorization of

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 2 & 2 \\ 3 & 6 & 4 \end{bmatrix}.$$

Is $LU = A$?

By swapping R1 and R3, then applying $R2 \rightarrow R2 - 2R1/3$ and $R3 \rightarrow R3 - R1/3$ then $R3 \rightarrow R3 - R1/2$

$$\begin{bmatrix} 3 & 6 & 4 \\ 2 & 2 & 2 \\ 1 & 1 & 3 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 6 & 4 \\ 0 & 2 & -2/3 \\ 0 & -1 & 5/3 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 6 & 4 \\ 0 & -2 & -2/3 \\ 0 & 0 & 2 \end{bmatrix} = U$$

and the lower triangular matrix is

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1/3 & 1/2 & 1 \end{bmatrix}$$

The MATLAB command `lu`

```
>> A = [1 0 3; 2 2 2; 3 6 -3];
```

```
>> [L1,U1,P] = lu(A)
```

L1 =

1.0000	0	0
0.6667	1.0000	0
0.3333	1.0000	1.0000

U1 =

3	6	-3
0	-2	4
0	0	0

P =

0	0	1
0	1	0
1	0	0

```
>> [L2,U2] = lu(A)
L2 =
    0.3333    1.0000    1.0000
    0.6667    1.0000         0
    1.0000         0         0
U2 =
     3     6    -3
     0    -2     4
     0     0     0
```

Note that L_2 is not a unit lower triangular matrix. It is a “psychologically lower triangular matrix” (i.e. a product of lower triangular and permutation matrices) in L_1 , so that $A = L_2 * U_2$.

Instability with well-conditioned matrices – Another reason for pivoting

Example 37

Consider the linear system

$$\begin{bmatrix} \mu & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

where $|\mu|$ is very small.

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- The exact solution is

$$\mathbf{x} = \frac{1}{1 - \mu} \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

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- $\kappa_{\infty}(A) = 4/|\mu - 1| \approx 4$, so the problem is well-conditioned.

Example 37 (cont.)

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- The LU factorization is

$$L = \begin{bmatrix} 1 & 0 \\ \mu^{-1} & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} \mu & 1 \\ 0 & 1 - \mu^{-1} \end{bmatrix},$$

Example 37 (cont.)

- The LU factorization is

$$L = \begin{bmatrix} 1 & 0 \\ \mu^{-1} & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} \mu & 1 \\ 0 & 1 - \mu^{-1} \end{bmatrix},$$

- Run the matlab script `badpivot.m` Interpret the outputs.

Example 37 (cont.)

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- This loss of accuracy is easily avoided. We simply change the order of the equations, i.e., we solve the equivalent system

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- The solution $\mathbf{x}$ and condition number $\kappa(\mathbf{A})$ are unchanged, but now

$$L = \begin{bmatrix} 1 & 0 \\ \mu & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 1 \\ 0 & 1 - \mu \end{bmatrix}.$$

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$$L = \begin{bmatrix} 1 & 0 \\ \mu & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 1 \\ 0 & 1 - \mu \end{bmatrix}.$$

- Now $\mathbf{x}_*$ will be accurate to machine precision.

Roundoff and LU factorization

Theorem 38 (Wilkinson 1961)

Suppose that the LU -factorization of a matrix A is computed in a system of floating-point arithmetic. Then the approximate quantities L_ and U_* satisfy*

$$L_* U_* = A + \delta A,$$

where

$$\|\delta A\|_\infty \leq \rho_n (\|A\|_\infty + \|L_*\|_\infty \|U_*\|_\infty) \mathbf{u} + O(\mathbf{u}^2).$$

Here, ρ_n is a constant not larger than $3n$ and $\mathbf{u} = \epsilon/2 \approx 1 \times 10^{-16}$.

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Here, ρ_n is a constant not larger than $3n$ and $\mathbf{u} = \epsilon/2 \approx 1 \times 10^{-16}$.

Typically, $\rho_n \ll n$ when n is large.

Roundoff and error in solving with LU factorization

The following theorem is a direct consequence of Theorems 18 and 38.

Theorem 39

Suppose that we solve $A\mathbf{x} = \mathbf{b}$ by finding the LU -decomposition $A = LU$ and then solving the triangular systems $L\mathbf{y} = \mathbf{b}$ and $U\mathbf{x} = \mathbf{y}$. Assume that we perform all computations in a system of floating-point arithmetic. Then the computed solution $\mathbf{x}_$ is the exact solution of a perturbed linear system*

$$(A + \delta A)\mathbf{x}_* = \mathbf{b} + \delta \mathbf{b},$$

and satisfies

$$\frac{\|\mathbf{x}_* - \mathbf{x}\|_\infty}{\|\mathbf{x}\|_\infty} \leq c_n \kappa_\infty(A) \left[\left(1 + \frac{\|L_*\|_\infty \|U_*\|_\infty}{\|A\|_\infty} \right) u + O(u^2) + \frac{\|\delta \mathbf{b}\|_\infty}{\|\mathbf{b}\|_\infty} \right].$$

Here c_n is a constant not larger than $5n$.

Remarks

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 - 2 All entries of the computed factors L_* and U_* are of moderate size.

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- Roundoff should not be a problem provided that the following two conditions hold:
 - 1 The condition number $\kappa_\infty(A)$ is of moderate size.
 - 2 All entries of the computed factors L_* and U_* are of moderate size.
- In Example 37, the condition number $\kappa_\infty(A)$ is approximately 4, but

$$\|L_*\|_\infty \approx \|L\|_\infty = 1 + |\mu^{-1}| \quad \text{and} \quad \|U_*\|_\infty \approx \|U\|_\infty = |1 - \mu^{-1}|.$$

Hence

$$\frac{\|L_*\|_\infty \|U_*\|_\infty}{\|A\|_\infty} = \frac{(1 + |\mu^{-1}|)|1 - \mu^{-1}|}{2} \approx \frac{1}{2\mu^2},$$

which is large when $|\mu|$ is very small.

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which is large when $|\mu|$ is very small.

- A remedy is **partial pivoting**.

Partial pivoting

Partial pivoting

- It is possible to perform row swaps (even in the case of nonzero pivot elements) so that a pivot element is larger in magnitude than all other entries in the same column lying below it. This process is called **partial pivoting**, and it guarantees that

$$|\ell_{ij}| \leq 1 \quad \text{for all } 1 \leq j < i \leq n.$$

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$$|\ell_{ij}| \leq 1 \quad \text{for all } 1 \leq j < i \leq n.$$

- The MATLAB function **lu** performs this pivoting.

Example 40

We illustrate how pivoting is implemented by factoring the 4×4 matrix

$$A = \begin{bmatrix} 1 & 0 & -5 & 2 \\ 5 & 1 & 8 & 4 \\ -3 & 3 & 7 & 2 \\ 1 & -4 & 8 & 2 \end{bmatrix} \begin{array}{|c|} \hline 1 \\ 2 \\ 3 \\ 4 \\ \hline \end{array}.$$

We use the boxed column of numbers to keep track of the row swaps. To do the computations in Matlab, we start with

```
>> A = [1, 0, -5, 2; 5, 1, 8, 4; -3, 3, 7, 2; 1, -4, 8, 2]  
>> saveA = A
```

Example 40 (cont.)

At the first step, we find the entry in the first column that has the largest magnitude, in this case 5. To use this number as the pivot, we swap its row with row 1:

```
>> A([1 2], :) = A([2 1], :)
```

In this way, we obtain

$$\begin{bmatrix} 5 & 1 & 8 & 4 \\ 1 & 0 & -5 & 2 \\ -3 & 3 & 7 & 2 \\ 1 & -4 & 8 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 3 \\ 4 \end{bmatrix}.$$

Example 40 (cont.)

Now eliminate the $(2, 1)$ -entry by subtracting $1/5 \times \text{row 1}$ from row 2, but instead of storing 0 in the $(2, 1)$ -position, store the **multiplier** $1/5 = 0.2$. The commands

```
>> A(1,2) = A(1,2) / A(1,1)
>> A(2,2:4) = A(2,2:4) - A(2,1) * A(1,2:4)
```

give

$$\begin{bmatrix} \mathbf{5.0} & 1.0 & 8.0 & 4.0 \\ \mathbf{0.2} & -0.2 & -6.6 & 1.2 \\ -3.0 & 3.0 & 7.0 & 2.0 \\ 1.0 & -4.0 & 8.0 & 2.0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 3 \\ 4 \end{bmatrix}.$$

Example 40 (cont.)

To eliminate the (3, 1)- and (4, 1)-entries (and store the multipliers), we do

```
>> A(3,1)      = A(3,1)/A(1,1)
>> A(3,2:4)    = A(3,2:4) - A(3,1)*A(1,2:4)
>> A(4,1)      = A(4,1)/A(1,1)
>> A(4,2:4)    = A(4,2:4) - A(4,1)*A(1,2:4)
```

and obtain

$$\begin{bmatrix} 5.0 & 1.0 & 8.0 & 4.0 \\ 0.2 & -0.2 & -6.6 & 1.2 \\ -0.6 & 3.6 & 11.8 & 4.4 \\ 0.2 & -4.2 & 6.4 & 1.2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 3 \\ 4 \end{bmatrix}.$$

Example 40 (cont.)

Looking at the entries in positions 2 to 4 of the second column, we see that the largest in magnitude is -4.2 , so we make this the next pivot by swapping rows 2 and 4.

```
>> A([2 4], :) = A([4 2], :)
```

In this way, we have

$$\begin{bmatrix} 5.0 & 1.0 & 8.0 & 4.0 \\ 0.2 & -4.2 & 6.4 & 1.2 \\ -0.6 & 3.6 & 11.8 & 4.4 \\ 0.2 & -0.2 & -6.6 & 1.2 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 3 \\ 1 \end{bmatrix}.$$

Example 40 (cont.)

Now eliminate the (3, 2)- and (4, 2)-entries, storing the multipliers:

$$>> A(3, 2) = A(3, 2) / A(2, 2)$$

$$>> A(3, 3:4) = A(3, 3:4) - A(3, 2) * A(2, 3:4)$$

$$>> A(4, 2) = A(4, 2) / A(2, 2)$$

$$>> A(4, 3:4) = A(4, 3:4) - A(4, 2) * A(2, 3:4)$$

and obtain

$$\begin{bmatrix} 5.0000 & 1.0000 & 8.0000 & 4.0000 \\ 0.2000 & -4.2000 & 6.4000 & 1.2000 \\ -0.6000 & -0.8571 & 17.2857 & 5.4286 \\ 0.2000 & 0.0476 & -6.9048 & 1.1429 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 3 \\ 1 \end{bmatrix}.$$

Example 40 (cont.)

Since the (3,3)-entry is larger in magnitude than the (4,3)-entry, no row swap is needed and we do

$$>> A(4,3) = A(4,3) / A(3,3)$$

$$>> A(4,4) = A(4,4) - A(4,3) * A(3,4)$$

to complete the factorization:

$$\begin{bmatrix} 5.0000 & 1.0000 & 8.0000 & 4.0000 \\ 0.2000 & -4.2000 & 6.4000 & 1.2000 \\ -0.6000 & -0.8571 & 17.2857 & 5.4286 \\ 0.2000 & 0.0476 & -0.3994 & 3.3113 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 3 \\ 1 \end{bmatrix}.$$

Example 40 (cont.)

Indeed, the lower triangular factor is

$$L = \begin{bmatrix} 1.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.2000 & 1.0000 & 0.0000 & 0.0000 \\ -0.6000 & -0.8571 & 1.0000 & 0.0000 \\ 0.2000 & 0.0476 & -0.3994 & 1.0000 \end{bmatrix},$$

and the upper triangular factor is

$$U = \begin{bmatrix} 5.0000 & 1.0000 & 8.0000 & 4.0000 \\ 0.0000 & -4.2000 & 6.4000 & 1.2000 \\ 0.0000 & 0.0000 & 17.2857 & 5.4286 \\ 0.0000 & 0.0000 & 0.0000 & 3.3113 \end{bmatrix}.$$

Example 40 (cont.)

If we define the **permutation matrix**

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

then

$$PA = LU.$$

To check this, do

```
>> L = tril(A,-1) + eye(4,4), U = triu(A), A = saveA
>> P = eye(4,4), P = P([2 4 3 1], :)
>> norm(L*U-P*A, inf)
ans = 8.8818e-16
```

To solve $A\mathbf{x} = \mathbf{b}$, we solve the triangular systems $L\mathbf{y} = P\mathbf{b}$ and $U\mathbf{x} = \mathbf{y}$, so that

$$PA\mathbf{x} = LU\mathbf{x} = L\mathbf{y} = P\mathbf{b}.$$

Motivation

Example 41

Run the Matlab script `solvers.m`

Operation Counts

Definition 42 (Flop)

A **flop** is a **floating-point operation**, i.e., one of $a + b$ or ab where a and b are two floating point numbers.

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Example 43

To compute an **inner product**

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n x_i y_i$$

we use $2n$ flops.

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Example 43

To compute an **inner product**

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n x_i y_i$$

we use $2n$ flops.

A simple count give $2n - 1$ flops for an inner product, not $2n$. However, a program to perform this product usually starts with $s = 0$ and then $s = s + x_i y_i$ for $i = 1, \dots, n$. Therefore, there are $2n$ flops. Moreover, we are concerned with large n , so $2n - 1 \approx 2n$.

Counting flops in this way gives some indication of the **computational cost** of an algorithm.

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Computing L and U uses roughly $\frac{2n^3}{3}$ flops. We say that the complexity in finding the LU factorization is $O(n^3)$.

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Conclusion

When we solve a linear system by LU factorization, most of the time is spent to find the LU factorization. Therefore, when the same system is solved with different right hand sides, it is recommended that the factorization is carried once and stored for later use.

The Matlab script `timelu.m` estimates the exponent p such that the CPU time to solve $Ax = b$ is approximately proportional to n^p .

```
steps = 10; neqns = 500 * [1:steps]; tlu = zeros(1,steps);

fprintf('Timing LU factorization of an n-by-n matrix.\n\n')
fprintf('      n              seconds          exponent\n\n')
for k = 1:steps
    n = neqns(k);
    A = eye(n) + 0.5 * rand(n);
    start = cputime;
    [L,U] = lu(A);
    finish = cputime;
    tlu(k) = finish - start;
    if k == 1 | tlu(k-1) < eps
        fprintf('%8d    %12.4f\n', n, tlu(k))
    else
        p = log( tlu(k)/tlu(k-1) ) / log( neqns(k)/neqns(k-1));
        fprintf('%8d    %12.4f    %10.2f\n', n, tlu(k), p)
    end
end
end
```

On my PC (3.2GHz P4), I got the following output:

Timing LU factorization of an n -by- n matrix.

n	seconds	exponent
500	0.0400	
1000	0.2500	2.64
1500	0.7600	2.74
2000	1.7000	2.80
2500	3.1400	2.75
3000	5.2600	2.83
3500	8.1600	2.85
4000	12.0600	2.93
4500	16.6800	2.75
5000	22.3400	2.77

Symmetric and positive definite matrices

Definition 44 (Symmetric)

A matrix $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ is **symmetric** if $A^T = A$, i.e., if

$$a_{ji} = a_{ij} \quad \text{for all } i \text{ and } j.$$

Symmetric and positive definite matrices

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Definition 45 (Positive definiteness)

The matrix $A \in \mathbb{R}^{n \times n}$ is said to be **positive-definite** if

$$\mathbf{x}^T A \mathbf{x} = \mathbf{x} \cdot (A \mathbf{x}) > 0 \quad \text{for all } \mathbf{x} \in \mathbb{R}^n \text{ except } \mathbf{x} = \mathbf{0},$$

or in other words if

$$\sum_{i=1}^n \sum_{j=1}^n x_i a_{ij} x_j > 0 \quad \text{except when } x_1 = x_2 = \dots = x_n = 0.$$

Example 46

The matrix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

is positive definite. In fact,

$$\begin{aligned} \mathbf{x}^T A \mathbf{x} &= 2x_1^2 - 2x_1x_2 + 2x_2^2 - 2x_2x_3 + 2x_3^2 \\ &= x_1^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + x_3^2 \\ &> 0, \end{aligned}$$

if $\mathbf{x} = (x_1, x_2, x_3) \neq \mathbf{0}$.

Theorem 47 (A criterion for positive definiteness)

Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix. Then A is positive definite if and only if

$$\det(A_k) > 0 \quad \text{for all } k = 1, \dots, n,$$

where A_k is the principal matrix of order k of A .

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Example 48

Consider again the symmetric matrix in Example 46. Then

$$A_1 = [2], \quad A_2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad A_3 = A.$$

Since

$$\det(A_1) = 2 > 0, \quad \det(A_2) = 3 > 0, \quad \det(A_3) = 4 > 0,$$

A is positive definite.

Another criterion

Theorem 49

Let $A \in R^{n \times n}$ be a symmetric matrix. If all the eigenvalues of A are positive then A is positive definite.

The eigenvalues of a matrix can be found using the MATLAB command `eig`.

Cholesky Factorization

Theorem 50

*A matrix $A \in \mathbb{R}^{n \times n}$ is symmetric and positive-definite if and only if A has a **Cholesky factorization**, i.e.,*

$$A = R^T R$$

where

Cholesky Factorization

Theorem 50

A matrix $A \in \mathbb{R}^{n \times n}$ is symmetric and positive-definite if and only if A has a *Cholesky factorization*, i.e.,

$$A = R^T R$$

where

- 1 R is upper triangular, i.e., $r_{ij} = 0$ for $1 \leq j < i \leq n$;

Cholesky Factorization

Theorem 50

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where

- 1 R is upper triangular, i.e., $r_{ij} = 0$ for $1 \leq j < i \leq n$;
- 2 $r_{ii} > 0$ for $i = 1, 2, \dots, n$.

Example 51

If (see Example 46)

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

then $A = R^T R$ where

$$R = \begin{bmatrix} \sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & \sqrt{3/2} & -\sqrt{2/3} \\ 0 & 0 & 2/\sqrt{3} \end{bmatrix}.$$

Example 51

If (see Example 46)

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

then $A = R^T R$ where

$$R = \begin{bmatrix} \sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & \sqrt{3/2} & -\sqrt{2/3} \\ 0 & 0 & 2/\sqrt{3} \end{bmatrix}.$$

`chol`

The MATLAB built-in function `chol` produces this factorization

$$R = \text{chol}(A)$$

Computational cost

The number of flops to compute R is about $\frac{n^3}{3}$, i.e., half the cost of an LU -factorization.

Computational cost

The number of flops to compute R is about $\frac{n^3}{3}$, i.e., half the cost of an LU -factorization.

Once R is found, we can solve

$$A\mathbf{x} = \mathbf{b}$$

at a cost of about n^2 flops by solving the triangular systems $R^T\mathbf{y} = \mathbf{b}$ and $R\mathbf{x} = \mathbf{y}$, so that

$$A\mathbf{x} = R^T R\mathbf{x} = R^T\mathbf{y} = \mathbf{b}.$$

Stability

Wilkinson showed that the relative rounding error in the computed solution satisfies

$$\frac{\|\mathbf{x}_* - \mathbf{x}\|_2}{\|\mathbf{x}\|_2} \leq \rho_n \kappa_2(\mathbf{A}) \epsilon,$$

where ρ_n is given in Theorem 38.